pyMETRIC-UAV: An adaptation of the METRIC surface-energy-balance model for evapotranspiration estimation from UAV thermal and multispectral imagery

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Abstract

pyMETRIC-UAV is a Python implementation of the METRIC (Mapping EvapoTranspiration at high Resolution with Internalized Calibration) model, adapted for sub-meter unmanned-aerial-vehicle (UAV) thermal and multispectral imagery. It is derived from the `hectornieto/pyMETRIC` repository and provides a flexible approach for identification of hot and cold pixels, along with an internal NDVI and LAI calibration that couples the radiometric end points for each to the same hot/cold surfaces used for the internal energy-balance calibration. The code base includes a streamlined batch workflow with explicit reference-surface, soil-heat-flux, and per-pixel fractional-cover handling. This document describes the theoretical background, the software architecture, the configuration system, and a full worked example for an image collected at the Colorado State Univeristy Western Colorado Reserach Center – Grand Valley Experimental Station located in Fruita, Colorado.

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1 Introduction

1.1 Evapotranspiration in agricultural water management

Evapotranspiration (ET) is the dominant outflow of water from cropped land, accounting for a majority of the water balance in irrigated agriculture (Allen et al. 1998). Quantifying ET at field scale is central to irrigation scheduling,drought response, and the evaluation of deficit-irrigation strategies. Direct measurement of ET from eddy-covariance towers, lysimeters, sap-flow probes and other methods can yield high-quality data. However, these methods are expensive, spatially sparse, and difficult to scale across heterogeneous agricultural landscapes. Remote-sensing approaches help close this gap by combining a satellite or aerial observations of the field with meteorological inputs and physical models of the surface energy balance.

1.2 Thermal surface-energy-balance models

Several families of remote-sensing ET algorithms have emerged in recent years. The *surface-energy-balance* (SEB) family of models solves the residual of the energy budget at the land surface:

$$R_n = G + H + LE \tag{1}$$

where R_n is net radiation, G is soil heat flux, H is sensible heat flux, and LE is latent heat flux (the energy equivalent of ET). Remote-sensing observations can provide land surface temperature (LST) and surface albedo, from which R_n, G, and H can be parameterized; LE is then recovered as the residual. Two widely used energy balance model implementations are SEBAL (Bastiaanssen et al. 1998) and METRIC (Allen, Tasumi, and Trezza 2007; Allen et al. 2007). Both rely on identifying two anchor pixels — a *cold* pixel representing well-watered vegetation with near-potential ET, and a *hot* pixel representing dry bare soil with near-zero ET — and using them as endmembers to internally calibrate the linear relationship between LST and the air-land surface temperature gradient that drives sensible heat flux. A related two-source approach, TSEB (Norman, Kustas, and Humes 1995; Kustas and Norman 1999), partitions energy fluxes between soil and canopy sub-elements rather than treating each pixel as a bulk surface. Simpler operational algorithms in the SSEBop family (Senay et al. 2013) sidestep the anchor-pixel calibration entirely at the cost of stronger reliance on gridded reference-ET inputs. The energy balance model implementation presented here modifies a previous implementation of the METRIC model.

1.3 The benefits of internal calibration

Both SEBAL and METRIC rely on the assumption that the slope and intercept of the dT–T_s relationship — where T_s is the surface temperature, T_a is the air temperature, and $dT = T_s - T_a$ is the gradient that drives sensible heat flux at the surface roughness height — can be *inferred* from two carefully selected pixels in the image itself. At a selected cold pixel, LE is assumed equal to a reference ET multiplied by a factor near unity, and H is the residual of Equation 1. At a selected hot pixel, LE is assumed equal to a small or zero ET fraction (the "ETrF_bare" term in Section 4), and H is again the residual. Two endmember values of dT, paired with their corresponding $T_{s,\text{datum}}$ values, define a linear calibration (Allen et al. 2007, Eq. 29):

$$dT = a + b \cdot T_{s,\text{datum}} \tag{2}$$

where $T_{s,\text{datum}}$ is the surface temperature normalized to a common elevation datum (defined in Equation 7 below). This calibration absorbs systematic biases in aerodynamic resistance, roughness parameterization, and air-temperature interpolation that would otherwise dominate the uncertainty in image-wide H, and reduces sensitivity to errors in any single ancillary input (Allen, Tasumi, and Trezza 2007; Allen et al. 2013; Long and Singh 2013).

1.4 Differences between satellite and UAV-scale implementations

The original implementation of METRIC was developed for Landsat-class imagery, with ~30 m thermal pixels and frequent revisits over agricultural landscapes (Allen et al. 2007). UAV-mounted thermal cameras deliver sub-meter resolution at the cost of small field of view and short acquisition windows but are valuable for:

- diagnosing within-field variability driven by irrigation system performance and local topography;
- separating crop water use from soil evaporation in row crops, orchards, and vineyards (Xia et al. 2016; Nassar et al. 2020);
- isolating ET signal at the plot scale for agronomic experiments where each treatment cell occupies tens of square meters;
- supporting precision irrigation prescriptions where the management unit is smaller than a Landsat pixel (Maes and Steppe 2019).

The same physics applies equally to satellite imagery and UAV imagery. However, several practical assumptions made in the satellite implementation of energy balance models break down for UAV imagery. UAV scenes may be collected over small areas that make identification of truly bare-and-dry hot pixels challenging or impossible. Similar challenges may exist for the identification of fully-watered cold pixels. Raw thermal imagery from UAV platforms may contain sensor artifacts (specular reflections, cold-edge halos) that satellite algorithm filters are not designed to catch. The UAV image field-of-view often spans only a single land-cover class, so the land cover-class-driven branches of the original METRIC algorithm collapse to a single pass. These challenges motivate the modifications described in Section 3.

1.5 Software lineage and scope

pyMETRIC-UAV is a fork of `hectornieto/pyMETRIC` (Nieto, Guzinski, and contributors 2018--), which is built on the pyTSEB family of models (Nieto and contributors 2017). We preserve the core energy-balance solver from pyMETRIC, the Monin-Obukhov stability iteration, and the integration with ASCE-standard reference-ET calculations. We replace the per-class land cover loop with a single-pass model, add an explicit reference-surface abstraction, introduce a four-mode endmember system with zone-polygon support, derive all model inputs from a single multiband image,include methods to sanitize UAV image artifacts, and produce a notebook-driven batch workflow. This document describes the resulting system as it stands at the time of writing.

2 Theoretical background

2.1 The surface energy balance

At each pixel, instantaneous fluxes are required to balance:

$$R_n - G - H - LE = 0 \tag{3}$$

with sign convention: R_n positive toward the surface (downwards), G positive into the soil, H and LE positive toward the atmosphere (away from the surface). All terms have units of W m^{-2}. Net radiation decomposes into shortwave and longwave components. The **conceptual** form is

$$R_n = R_{ns} + R_{nl} = (1 - \alpha)S_{\downarrow} + \varepsilon\, L_{\downarrow} - \varepsilon\, \sigma T_s^4 \tag{4}$$

where α is broadband shortwave albedo, $S_{\downarrow}$ is incoming shortwave radiation (a meteorological input), ε is broadband surface emissivity, $L_{\downarrow}$ is incoming longwave radiation (measured or estimated from T_a and e_a via an atmospheric emissivity parameterization), and σ is the Stefan–Boltzmann constant.

In `pyMETRIC-UAV` the longwave term $R_{nl} = \varepsilon\, L_{\downarrow} - \varepsilon\, \sigma T_s^4$ is computed exactly as written (METRIC.py:194). The shortwave term, however, is **not** computed as the bulk product $(1 - \alpha)\, S_{\downarrow}$. Instead, $S_{\downarrow}$ is split into direct and diffuse components and a Campbell two-layer radiative transfer model (Campbell 1986) is applied separately to the canopy and soil sub-elements (`pyTSEB.net_radiation.calc_Sn_Campbell`), driven by LAI, solar zenith angle, leaf optics ($\rho_{\mathrm{vis,C}}, \tau_{\mathrm{vis,C}}, \rho_{\mathrm{nir,C}}, \tau_{\mathrm{nir,C}}, x_{\mathrm{LAD}}$), and soil optics ($\rho_{\mathrm{vis,S}}, \rho_{\mathrm{nir,S}}$). The per-pixel albedo is *back-calculated* from the result as $\alpha = 1 - R_{ns}/S_{\downarrow}$. Equation 4 should be read as the energy-balance partition that R_n ultimately satisfies, not as the formula the code evaluates.

2.2 Soil heat flux

Soil heat flux is the most weakly constrained term in the budget. pyMETRIC offers five parameterizations, selected by the `G_form` configuration parameter:

G_form	Formulation	Inputs
0	Constant value	`G_constant` (W m^{-2})
1	Fixed ratio of R_n	`G_ratio` (unitless)
2	(Santanello and Friedl 2003)	`G_amp`, `G_phase`, `G_shape`
4	ASCE reference-ET ratio	`G_tall` (alfalfa) or `G_short` (grass)
5	(Allen, Tasumi, and Trezza 2007)	per-pixel from T_s, α, NDVI

Option 4 is the default and uses one ratio per scene, picked by the `reference_type` setting (see Section 2.5). Option 5 is recommended for mixed-cover UAV scenes (e.g. orchard and pasture) because it lets G/R_n respond to local canopy density without requiring a multi-class landcover loop.

2.3 Sensible heat flux

Sensible heat is computed by the bulk transfer equation:

$$H = \rho\, c_p\, \frac{dT}{r_{ah}} \tag{5}$$

where ρ is air density, c_p is the heat capacity of moist air, dT is the air-surface temperature gradient at the height where r_{ah} is evaluated, and r_{ah} is the aerodynamic resistance to heat transfer. Rather than measuring dT directly (which would require high-quality air-temperature data at two elevations–the ground surface and the top of the canopy–co-located with each pixel in the the thermal scene), or computing it from $T_s - T_a$ (which would require absolutely-accurate T_s), METRIC infers it from the linear relation:

$$dT = a + b \cdot T_{s,\text{datum}} \tag{6}$$

where $T_{s,\text{datum}}$ is the surface temperature adjusted to a common elevation datum:

$$T_{s,\text{datum}} = T_s + \gamma_w\, z \tag{7}$$

where γ_w is the moist-adiabatic lapse rate (computed from T_a, e_a, p) and z is the per-pixel elevation. In satellite scenes, the datum adjustment prevents elevation differences within a scene from biasing the calibration toward the colder high-elevation pixels. `pyMETRIC-UAV` treats each scene as sharing a common elevation, so the datum adjustment in Equation 7 reduces to a constant offset that cancels out of the calibration; the implementation therefore omits it and calibrates dT directly against the raw surface temperature ($dT = a + b\, T_s$). The coefficients a and b are then solved at the two endmember pixels by the simultaneous system of equations:

$$\begin{aligned} H_{\text{cold}} &= R_{n,\text{cold}} - G_{\text{cold}} - LE_{\text{cold}} \\ H_{\text{hot}} &= R_{n,\text{hot}} - G_{\text{hot}} - LE_{\text{hot}} \end{aligned} \tag{8}$$

where LE_{cold} is the cold-pixel reference-ET fraction ($\text{ETrF}_{\text{cold}} \approx 1.05$) times an ASCE-standard reference ET, and LE_{hot} is the hot-pixel ETrF (default 0 for fully dry surfaces; configurable via `ETrF_bare` or directly via `hot_ETrF`) times the same reference ET. Inverting Equation 5 at each endmember yields dT_{cold} and dT_{hot}, which are fitted to two points on the line derived by Equation 2. The resulting slope and intercept terms are then applied to each pixel in the image.

When the cold pixel value falls below the advective demand (e.g., $1.05\,ET_{ref} > R_n - G$), the residual H_{cold} will go negative, potentially introducing instability into the Monin–Obukhov iteration or pushing it toward a degenerate stable solution where r_{ah} inflates without bound and the value for dT_{cold} become physically unrealistic. Following Allen et al. (2013), `pyMETRIC-UAV` floors the cold-pixel residual at a small positive value ($H_{\text{cold,min}} = 5$ W m^{-2}); equivalently, LE_{cold} is capped at $R_n - G - H_{\text{cold,min}}$ when the calibrated value exceeds the available energy. A message is logged whenever the cap is applied.

2.4 Monin–Obukhov stability iteration

Aerodynamic resistance r_{ah} depends on near-surface turbulent mixing, which itself depends on the buoyancy of the surface layer through the Monin–Obukhov length:

$$L = -\frac{u_*^3\,\rho\,c_p\,T_a}{\kappa\,g\,(H + 0.61\,c_p\,T_a\,LE/\lambda)} \tag{9}$$

where u_* is friction velocity, κ is von Kármán's constant, g is gravitational acceleration, and λ is the latent heat of vaporization. The denominator includes the virtual-temperature buoyancy correction $0.61\,c_p\,T_a\,LE/\lambda$. Allen et al. (2007, Eq. 40) drops this term and uses H alone in the denominator, substituting T_s for T_a in the numerator. `pyMETRIC-UAV` inherits the full virtual-temperature form from pyTSEB (Nieto and contributors 2017), which is slightly more general but typically agrees with the Allen (2007) simplification to within a few percent when LE and H are of similar magnitude over irrigated surfaces.

L enters the stability-corrected wind and temperature profiles that determine r_{ah}, but L itself depends on H and LE, which in turn depend on r_{ah} via Equation 5. The system is solved iteratively following the steps below:

1. Assume neutral atmospheric conditions ($L \to \pm\infty$).
2. Compute r_{ah} and u_* from the stability-corrected profiles.
3. Compute H from Equation 5 using the current dT.
4. Compute LE as the residual of Equation 3.
5. Update L from Equation 9.
6. Repeat until L converges below a relative threshold (10^{-3} in our implementation).

The endmember calibration of Section 2.3 is performed first, using its own short Monin–Obukhov loop at the two anchor pixels. The image-wide solution then proceeds with the calibrated (a, b). The code also detects and breaks out of oscillating limit cycles (queue of length = 6) that occasionally appear at very stable or very unstable pixels.

2.5 Reference ET formulation

The ASCE standardized reference ET for hourly time steps is

$$ET_{sz} = \frac{0.408\,\Delta(R_n - G) + \gamma\frac{C_n}{T+273}\,u_2\,(e_s - e_a)}{\Delta + \gamma(1 + C_d\,u_2)} \tag{10}$$

where Δ is the slope of the saturation-vapor-pressure curve (kPa °C^{-1}), γ is the psychrometric constant (kPa °C^{-1}), T is mean hourly air temperature (°C), u_2 is wind speed 2 m above the ground surface (m s^{-1}, log-profile-corrected from the measurement height), e_s and e_a are saturation and actual vapor pressures (kPa), and C_n, C_d are reference-specific constants given below.

The endmember calibration anchors LE_{cold} at a small multiple of the *reference* ET — the ET that a hypothetical well-watered reference surface would experience under the same meteorological conditions. The ASCE-EWRI standard (ASCE-EWRI 2005) defines two reference surfaces:

- ET_r: *Tall* reference (alfalfa-like), 0.5 m canopy height, $C_n = 66$, $C_d = 0.25$, daytime $G/R_n = 0.04$.
- ET_o: *Short* reference (clipped-grass), 0.12 m canopy height, $C_n = 37$, $C_d = 0.24$, daytime $G/R_n = 0.10$.

ET_r is typically 20 to 30% greater than ET_o under daytime conditions (Allen, Tasumi, and Trezza 2007). The choice of reference surface propagates through the model in three places:

1. **Pixel-wise** ET_{ref} (ancillary output bands 7–8) and the *ET fraction* $fETr = LE/ET_{ref}$ (ancillary output band 9).
2. **Cold-pixel anchor** LE_{cold}, hence the calibrated dT intercept a and slope b.
3. **Soil heat flux when `G_form=4`**, which uses `G_tall` or `G_short` accordingly.

The default configuration in `pyMETRIC-UAV` is ET_r (`tall`), matching standard ASCE practice for irrigated agriculture in the western U.S. The reference ET equation is used in two places. The function `batch_helpers.calc_etr_inst` computes a single point-value reference ET written to `*_Penman_ET.csv`. The model function `pyMETRIC.METRIC.pet_asce` computes the ET_{ref} band and the cold-pixel anchor LE_{cold}. Both functions read the same `reference_type` so the constants C_n and C_d stay in sync.

Both the notebook helper (`batch_helpers.calc_etr_inst`) and the model implementation (`pyMETRIC.METRIC.pet_asce`) use the full ASCE-EWRI solution. The version of `pet_asce` inherited from `hectornieto/pyMETRIC` was missing the `1 +` term in the denominator and biased ET_{ref} and the cold-pixel anchor upward by ~30–60 % at typical UAV wind speeds. The solutions produced by the full ASCE-EWRI f_{cd} parameterization in the notebook helper vs. the surface-balance `calc_Ln` form inside the model agree to within the small differences arising from how each estimates net longwave.

2.6 LAI and fractional cover computations

Where direct LAI measurements are not available, `pyMETRIC-UAV` derives fractional vegetation cover from NDVI using the squared-NDVI form after Carlson and Ripley (1997):

$$f_c = \left(\frac{NDVI - NDVI_{\text{soil}}}{NDVI_{\text{full}} - NDVI_{\text{soil}}} \right)^2 \tag{11}$$

LAI is then recovered from f_c via Beer's-law inversion (Choudhury 1987):

$$LAI = -\frac{\ln(1 - f_c)}{k_{\text{ext}}} \tag{12}$$

The canonical METRIC implementation (Allen, Tasumi, and Trezza 2007) derives LAI directly from SAVI rather than NDVI (see their Eq. 18). We use the NDVI route because it requires one fewer parameter and matches the parameters commonly avaiable from UAV multispectral images. The two endpoint parameters $NDVI_{\text{soil}}$ and $NDVI_{\text{full}}$ nominally come from a per-crop look-up. However, when zone polygons are provided for hot and cold pixel identification (see Section 4.4) we compute the median NDVI inside the hot and cold polygons, respectively. This couples the LAI derivation to the same physical surfaces that anchor the energy-balance calibration. A fully-watered surface defines the maximum NDVI while a bare-soil surface defines the minimum. This approach removes one source of free parameters from crop specific calculations. The extinction coefficient k_{ext} remains a crop-specific input, typically in the range 0.45–0.65.

3 Software architecture

3.1 Package layout

The repository is organised into two source layers and a workflow notebook:

```
pyMETRIC-batch/
  pyMETRIC/
     METRIC.py                       energy-balance core (METRIC, pet_asce)
     PyMETRIC.py                     image-processing wrapper (PyMETRIC class)
     endmember_search.py             hot/cold pixel algorithms + zone masks
     METRICConfigFileInterface.py config parser, file auto-discovery
```

```
    batch_helpers.py             notebook-side support (preflight,
                                  derivation, Sanitization, ET conversion,
                                  reference ET, zonal stats, diagnostics)
  pyMETRIC_Batch_Run.ipynb        six-cell batch workflow notebook
  config_template.txt             annotated per-dataset config
  METRIC_local_image_main.py      CLI entry point for processing a single image
  Batch_Data/                     contains one <SiteName> folder per dataset
      <SiteName>/
          config.txt
          <multiband_image>.tif
          Input/                  auto-created
              NDVI.tif / LAI_NDVI.tif / FC.tif / TRAD.tif
              endmember_zones.gpkg (user-provided)
              study_plots.gpkg     (user-provided, optional)
          Output/                 auto-created
              <SiteName>_METRIC.tif           (Rn, H, LE, G)
              <SiteName>_METRIC_ancillary.tif (9 ancillary bands)
              <SiteName>_ET_mm_hour.tif
              <SiteName>_Penman_ET.csv
              <SiteName>_Zonal_Stats.csv
              <SiteName>_log.txt
```

The `pyMETRIC/` package is what `import pyMETRIC` exposes to user code. The notebook deploys this directory over any instance of pyMETRIC installed in site-packages before importing, so a single source of truth is always in force. `batch_helpers.py` consolidates the workflow helpers, keeping the notebook itself to six short cells. Figure 1 shows the high-level data flow.

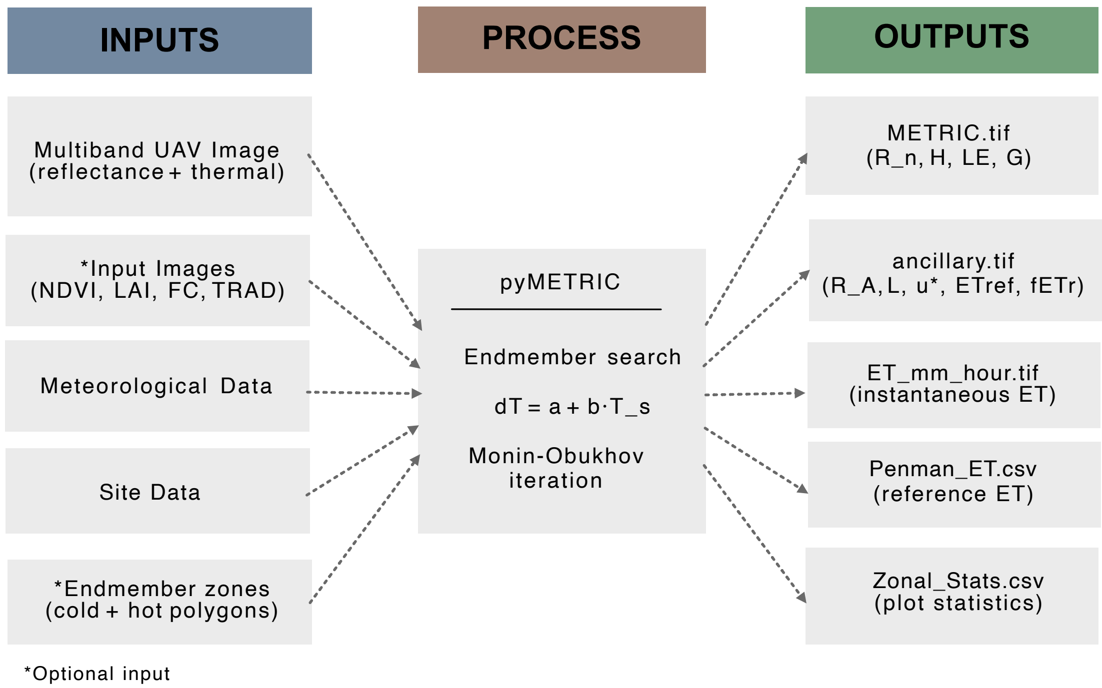

Figure 1: pyMETRIC-UAV workflow. The multiband UAV image, met values, site geometry, and (optionally) endmember zone polygons feed the model; outputs are written as a primary 4-band raster, a 9-band ancillary raster, a derived ET raster, a one-row reference-ET CSV, and a plot-level zonal-statistics CSV.

4 Configuration

4.1 Structure

A `pyMETRIC-UAV` config file is a flat key-value text file with `#` comments. The template (`config_template.txt`) is organised into three sections ordered by edit frequency:

Section	Purpose	Typical edit cadence
1. Flight info	Input image + sensor band assignments (1.1); date/time (1.2); meteorology (1.3); polygon files (1.4); endmember search (1.5)	every flight
2. Site info	lat, lon, alt, stdlon, instrument heights, `crop_type`, `reference_type`	each new site
3. Advanced settings	Crop presets (3.1), site-level canopy/leaf defaults (3.2), soil heat flux (3.3), QC (3.4), soil optics (3.5), aerodynamic resistance (3.6), optional file overrides (3.7), hot-pixel ETrF tuning (3.8), alternate endmember specifications (3.9)	rarely

A typical user will revise Section 1 for every flight image. This section gathers everything that is specific to an image dataset. Most users will adjust Section 2 once for a collection of images collected at the same locations. Values in Section 3 generally will not require user revision. Most users should be able to produce a working config by editing roughly 15 numeric values from a template file.

4.2 Crop presets

The active crop preset is selected with a single line in Section 2:

```
crop_type=alfalfa
```

Five canned presets are bundled with the template:

`crop_type`	k_{ext}	h_C (m)	$NDVI_{\text{full}}$	landcover (IGBP)
`alfalfa` (default)	0.60	0.5	0.90	12 (Crop)
`corn` (maize)	0.45	2.0	0.92	12 (Crop)
`wheat`	0.50	0.9	0.90	12 (Crop)
`pasture` (grass hay)	0.55	0.15	0.85	10 (Grass)
`orchard` (deciduous tree)	0.45	3.5	0.85	4 (Broadleaf D.)

Each preset block in Section 3.1 is defined with a dotted-prefix `<crop>.<param>` syntax:

```
alfalfa.k_ext=0.6
alfalfa.h_C=0.5
alfalfa.landcover=12
...
corn.k_ext=0.45
corn.h_C=2.0
corn.landcover=12
...
```

Adding a new crop is a copy-and-rename of any existing block plus one edit in Section 2. Tuning the preset values for a specific crop type requires a direct edit inside the prefixed block (e.g. `alfalfa.h_C=0.4` for a shorter alfalfa stand). Leaf optical properties (`rho_vis_C`, `tau_vis_C`, `rho_nir_C`, `tau_nir_C`) are defined in each crop type block. Properties that don't depend on crop — leaf emissivity (`emis_C`), canopy width:height ratio (`w_C`), leaf angle distribution (`x_LAD`), `NDVI_soil`, `LAI_max` — live as flat keys in Section 3.2. Soil optical properties (`rho_vis_S`, `rho_nir_S`, `emis_S`) are site-dependent rather than crop-dependent and live in Section 3.5. Preset values are intended as starting points only. Users with measured canopy data are encouraged to override the preset values.

4.3 The single-pass model

The upstream `hectornieto/pyMETRIC` implementation was designed for satellite-scale Landsat scenes spanning many International Geosphere-Biosphere Programme (IGBP) landcover classes. Its `run_METRIC` method looped twice over the image, once for IGBP classes corresponding to "tall" vegetation (forests, woodland savannas) and once for "short" vegetation (grass, crops, bare soil). The code silently picked both the reference-ET setting (alfalfa vs. grass) and the G/R_n ratio. For UAV scenes with a scalar landcover, overlapping landcover classes provided in `CROP_MOSAIC` could cause some pixels to be processed twice, with the second pass overwriting the first. The emergent reference-surface selection was then inconsistent with the implementation discussed in Section 2.5. The current implementation collapses the landcover algorithm into a single pass with the following steps:

1. Build an AOI mask that includes all valid pixels (i.e. not water/urban/snow).
2. Resolve `reference_type` once, deriving the ET_{ref} and the active G ratio.
3. Compute ET_{ref} and $ET_{ref,\mathrm{datum}}$ across the AOI.
4. Run the configured endmember mode (see Section 4.4).
5. Solve r_{ah}, H, LE, L, u_* iteratively for the whole AOI.

Landcover continues to flow through `pyTSEB.resistances.calc_roughness` for per-pixel z_{0m} and zero-plane displacement, so a heterogeneous landcover raster still yields heterogeneous roughness. However, the landcover class-conditional reference and G choices were removed.

4.4 Endmember selection

UAV scenes do not always contain an ideal hot or cold pixel. To accommodate the various cases likely encountered in UAV images, `pyMETRIC-UAV` exposes four endmember selection modes (`endmember_mode`):

`auto` Automatic search across the full valid AOI. The active algorithm is selected by `endmember_search` (CIMEC, ESA, or Min/Max). This is the default but, potentially, least robust for narrow scenes that lack a clear hot or cold surface.

`zone` Automatic search restricted to user-defined polygons (or bounding boxes). The polygon file is a GeoPackage (`endmember_zones.gpkg`) or Shapefile with at least two polygon features, each labelled `cold` or `hot` (case-insensitive, matched against the `zone` attribute by default, or a user-specified `endmember_zone_field`)

`manual` Endmember coordinates supplied directly. Accepts either pixel indices (`cold_pixel_rc=row,col`, `hot_pixel_rc=row,col`) or georeferenced points (`cold_pixel_xy=x,y`, `hot_pixel_xy=x,y`). Potentially useful when data from a flux tower or lysimeter provides information at a known location.

`prescribed` The dT calibration is bypassed entirely. The user supplies `prescribed_dT_a` and `prescribed_dT_b` directly, typically derived from a prior calibrated run or from independent flux-tower energy-balance closure. The hot- and cold-pixel logic is skipped and no `LE_hot` is computed.

In `auto` and `zone` modes, the user separately picks the search algorithm (`endmember_search`):

`0 = CIMEC` The Calibration using Inverse Modelling at Extreme Conditions algorithm from Allen et al. (2013). Filters by percentile and spatial homogeneity. Robust against single outlier pixels but assumes a sufficiently large candidate pool.

`1 = ESA` The Exhaustive Search Algorithm from Bhattarai et al. (2017). Histogram-trims outliers, then incrementally expands the search by percentile rank, then ranks candidates by VI vs LST opposition.

`2 = Min/Max` A simple percentile-based extreme selector. Picks the coldest 2% of vegetated pixels and the hottest 2% of bare pixels (both configurable), with the highest-VI / lowest-VI pixel chosen as the tie-break. Absolute min/max of LST computed on raw UAV thermal imagery may reflect sensor artifacts. The percentile-with-VI-tiebreak version is robust without losing the simplicity of the approach (see Section 6).

If provided by the user, the `endmember_zones.gpkg` polygons drive two independent computations:

1. **NDVI calibration** (before the model runs): the median NDVI inside the cold polygon sets $NDVI_{\text{full}}$ and the median inside the hot polygon sets $NDVI_{\text{soil}}$, which then feed the Beer's-law derivation of LAI and FC (Section 2.6).
2. **Endmember search** (in `zone` mode): the same polygons restrict the area in which the hot and cold pixels can be selected by the selected algorithm.

Enlarging the cold polygon raises $NDVI_{\text{full}}$ *and* broadens the cold-pixel search area. Both effects act to make the calibration "wetter." Polygons should be drawn tightly around physically representative surfaces — tens to a hundred square meters each is likely sufficient.

4.5 Sanity-checking via preflight

The notebook's preflight cell parses each config and emits human-readable warnings via `batch_helpers.sanity_checks`:

- The `DOY` and `time` values are converted to a calendar date for visual verification (a DOY typo is otherwise easy to miss).
- `stdlon` is compared against `lon` and flagged if the implied time zone is more than 15° away from the local meridian.

- Meteorologial values are compared against physically plausible ranges (T_A in [233, 333] K, u in [0, 25] m s^{-1}, $S_\downarrow$ in [0, 1400] W m^{-2}, p in [500, 1100] mbar).
- Inconsistent endmember-mode combinations (e.g. `endmember_zones` set with `endmember_mode=auto`) are noted.

These checks run before any heavy I/O and catch common configuration mistakes that would otherwise propagate silently into the energy balance solution.

4.6 Sanitization pipeline

Raw thermal imagery often contains outlier pixels that are physically impossible (e.g., sub-200 K values from missing-data substitution, NDVI beyond ±1 from numerical issues in the band ratio, negative LAI values from extrapolation outside the calibration range). The notebook includes a Sanitization block that replaces these pixels in place:

- **LAI**: negative values are clamped to 0; values above 10 (likely an artifact of FC values pushed against the upper clip in Equation 11) are capped to the user-configurable `LAI_max`.
- **TRAD**: values below 200 K are replaced with NaN.
- **NDVI**: values outside [-1, 1] are replaced with NaN.

The NDVI Sanitization runs after the derivation step, so the LAI and FC rasters that depend on NDVI are already consistent. The derivation step itself also masks pixels with zero or negative reflectance in either red or NIR (physically implausible) and propagates NaN through the NDVI–FC–LAI chain.

5 Inputs and outputs

5.1 The multiband UAV image

The default input is a single multiband GeoTIFF containing co-registered reflectance and thermal bands acquired during a single UAV flight. The config specifies which indexed bands hold red, NIR, and thermal data, along with the units of the thermal band. pyMETRIC-UAV makes no assumptions about additional bands, it reads only the three it needs.

5.2 Derived input rasters

The model run produces four GeoTIFFs that are written to the `Input/` folder:

File	Variable	Units	Range
`NDVI.tif`	VI	unitless	[-1, 1]
`LAI_NDVI.tif`	LAI	m^2 m^{-2}	[0, `LAI_max`]
`FC.tif`	f_c	unitless	[0.001, 0.99]
`TRAD.tif`	T_{R1}	K	[200, 350] typical

All four output files share the same spatial extent, resolution, and CRS as the source multiband image. They are written as Float32 with NaN as the nodata value.

5.3 File auto-discovery

If the configuration file does not explicitly set `T_R1`, `VI`, `LAI`, or `f_c`, the interface searches the `Input/` folder for files matching standard names (`TRAD.tif`, `NDVI.tif`, `LAI_NDVI.tif`, `FC.tif`, etc.) and sets the corresponding parameter automatically. This convention means that the notebook's derivation step (which writes `Input/NDVI.tif`, `LAI_NDVI.tif`, `FC.tif`, and `TRAD.tif`) hands off cleanly to the model without any path-management code in the configuration or notebook. Users can also drop pre-computed rasters into `Input/` for sites where the multiband-image flow doesn't apply, and the same model code consumes them. Path values are resolved relative to the configuration file's directory.

5.4 Primary output

The main output is a 4-band Float32 GeoTIFF (`*_METRIC.tif`) with the instantaneous energy-balance components in W m^{-2}:

Band	Variable	Units
1	R_n — net radiation	W m^{-2}
2	H — sensible heat flux	W m^{-2}
3	LE — latent heat flux	W m^{-2}
4	G — soil heat flux	W m^{-2}

The closure $R_n = H + LE + G$ holds exactly per pixel by construction (LE is the residual after the others are computed).

5.5 Ancillary output

The 9-band ancillary GeoTIFF (`*_METRIC_ancillary.tif`) carries diagnostic and intermediate variables:

Band	Variable	Units	Notes
1	R_{ns}	W m^{-2}	net shortwave
2	R_{nl}	W m^{-2}	net longwave
3	r_{ah}	s m^{-1}	aerodynamic resistance
4	L	m	Monin–Obukhov length
5	u_*	m s^{-1}	friction velocity
6	flag	–	0 = valid, 255 = invalid
7	$ET_{ref,\mathrm{datum}}$	W m^{-2}	reference ET (= band 8 with no delapse)
8	ET_{ref}	W m^{-2}	reference ET at the scene conditions
9	$fETr = LE/ET_{ref}$	–	ET fraction

`fETr` is the most useful single diagnostic for irrigation applications. A value near 1.0 indicates the surface is transpiring near its reference potential, and values close to zero indicate water stress or bare soil.

5.5.1 Instantaneous ET Output

The notebook's post-processing converts the model's LE output to instantaneous ET in mm hr^{-1} using the temperature-dependent latent heat of vaporisation:

$$\lambda = (2.501 - 0.002361\, T_C) \times 10^6 \quad [\mathrm{J\ kg^{-1}}] \tag{13}$$

$$ET = \frac{LE \cdot 3600}{\lambda} \quad [\mathrm{mm\ hr^{-1}}] \tag{14}$$

This single-band raster is written as `<SiteName>_ET_mm_hour.tif` and is the form most users actually plot, compare to reference ET, or (dis)aggregate to plot scale.

5.5.2 Reference ET Output

For each dataset the notebook writes a one-row CSV (`<SiteName>_Penman_ET.csv`) with the ASCE-EWRI reference ET computed from the met values in the config, using the same `reference_type` as the model. This is the value reported in summary tables and is directly comparable to the model's ancillary bands 7-8.

5.5.3 Zonal statistics

If `zonal_polygons` is set in the config, the notebook computes per-feature statistics (mean, median, min, max, standard deviation, pixel count, valid fraction) against the instantaneous ET raster and the ancillary $fETr$ band, and writes them as `<SiteName>_Zonal_Stats.csv`. Each zonal polygon is represented in a single row with `Variable` distinguishing `ET` from `fETr` values.

5.6 Log files

A processing log (`<SiteName>_log.txt`) captures the printed output of each pipeline stage — derivation, sanitization, METRIC execution, LE→ET conversion, reference ET, zonal stats. It is the first thing to inspect when a dataset produces unexpected results.

6 Sensitivity analysis

The notebook contains an optional sensitivity cell that re-runs each dataset under all six combinations of endmember algorithm × mode settings (CIMEC, ESA, Min/Max × auto, zone) and tabulates mean ET for each combination.

In a well-behaved scene, the legitimate spread between CIMEC, ESA, and Min/Max in zone mode reflects sensitivity to pixel selection by each algorithm within the same zone. Differences greater than 10–20% in image-wide mean ET may suggest one of:

- One or both zone polygons are too large, allowing algorithms to diverge on selected candidates.
- The selected hot zone exhibits insufficient bare-soil characteristics (i.e. NDVI is not low enough), causing the fallback "hottest pixel regardless of VI" branch to fire.
- A genuinely heterogeneous AOI where the assumption of a single (a, b) line is invalid.

7 Discussion and limitations

7.1 METRIC vs alternative models

METRIC's strength lies in its internal calibration, which absorbs systematic biases in the absolute scale of r_{ah}, z_{0m}, and air-temperature extrapolation that would otherwise dominate uncertainty. This makes it well-suited to operational mapping where the *pattern* of ET across a field matters as much or more than the absolute ET magnitude. For comparison:

- **SEBAL** (Bastiaanssen et al. 1998) uses essentially the same framework but without explicit reference-ET coupling; the slope and intercept of dT come from "wet" and "dry" pixels directly. METRIC's reference-ET anchoring is generally considered an improvement for irrigated systems where the ASCE-standard reference ET is well-defined.
- **SSEBop** (Senay et al. 2013) avoids the per-image endmember search by using *predefined boundary conditions unique to each pixel* for the hot and cold reference temperatures, then computing *ET* as a linear fraction of a gridded reference ET that scales with the LST anomaly against those per-pixel references. This approach is simpler operationally. There is no anchor-pixel search per scene. However, the references are not scene-adaptive, so the method can miss site-specific calibration that METRIC captures.

- **TSEB** (Norman, Kustas, and Humes 1995) partitions energy fluxes between soil and canopy sub-elements and does not require endmember calibration at all. It is preferable when the goal is to separately quantify soil evaporation and canopy transpiration. The trade-off is sensitivity to canopy fraction and roughness, which are themselves estimated.
- **eeMETRIC** (University of Idaho and Desert Research Institute 2018) is the Google Earth Engine implementation of the same algorithm family at Landsat scale, with a different endmember-search heuristic, automatic Landsat scene fetching, and a daily-integration step using ASCE reference ET. The two systems agree closely on Landsat scenes when fed the same met data. The primary practical difference from `pyMETRIC-UAV` is that the latter is designed for *single-acquisition* sub-meter UAV imagery and pays no attention to temporal integration.

7.2 pyMETRIC-UAV limitations

The package is deliberately scoped to single-flight, instantaneous ET. It does not provide:

- *Daily or seasonal integration* of instantaneous ET. The instantaneous $fETr$ band can be used externally with a daily reference-ET time series to estimate daily ET via the constant $fETr$ assumption (Tasumi et al. 2005), but this step is left to the user because it depends on additional meteorological data and assumptions about diurnal evolution of $fETr$ that are flight-specific.
- *Multi-flight time-series stitching.* Each dataset is treated as independent; there is no cross-flight calibration of dT coefficients, no temporal smoothing, no gap-filling.
- *Thermal-band atmospheric correction.* The model assumes the supplied TRAD raster is already in absolute land-surface temperature units. Raw thermal-camera digital counts must be radiometrically corrected and (where appropriate) atmospherically corrected upstream of this workflow.
- *Multi-source / two-source partitioning.* Energy fluxes are computed at the bulk pixel level, not partitioned between soil and canopy. Users interested in partitioning energy fluxes should explore the use of `pyTSEB`.

Other practical limitations included the following:

- **Mixed cover requires careful G choice.** `pyMETRIC-UAV` uses one G/R_n ratio per scene based on the selected `reference_type` by default. For mixed scenes (e.g. orchard and pasture), the uniform $G_{\text{tall}} = 0.04$ may underestimate G over the pasture by half. Use `G_form=5` (Allen 2007 empirical) for mixed scenes; G/R_n then varies per pixel based on local NDVI and albedo and tracks canopy density automatically.
- **Endmember zones are dual-purpose.** Drawing the cold polygon too large raises $NDVI_{\text{full}}$ (because it is computed on the the median NDVI inside the cold polygon) *and* broadens the cold-pixel search area. Both effects bias the calibration toward "wetter" values. Keep polygons tightly bound. Tens to a hundred square meter zones are likely sufficient in area.

- **Reference type must be picked consistently.** The `reference_type` setting drives the model's ET_{ref} band, the notebook's reported reference ET, and the active G ratio under `G_form=4`. The default `tall` (alfalfa) is appropriate for most western U.S. irrigated agriculture; switch to `short` (grass) if your reporting convention follows FAO-56 for non-alfalfa references.
- **Thermal sensor artifacts matter.** Single sub-200 K pixels in raw TRAD will be masked by the sanitizer, but artifacts in the 220–240 K range can pass through and affect the Min/Max algorithm if zone polygons are not used. The percentile-based replacement (Section 4.4) mitigates this but does not eliminate it; visual inspection of TRAD via the diagnostics cell remains valuable.

7.3 Possible future development directions

Useful extensions that are out of scope here but compatible with the current architecture:

- **Flux-tower validation of the corrected `pet_asce`.** The ASCE-EWRI denominator was restored in this version (see Section 2.5), and the Allen 2013 cold-pixel safeguard (floor on H_{cold}) was added to prevent advection-dominated cold pixels from driving the Monin–Obukhov iteration into a degenerate stable regime. Both changes shift absolute fluxes relative to earlier runs; cross-validation against eddy- covariance or against eeMETRIC on a co-located Landsat scene is the right next step before adopting these values for reporting beyond comparative analysis.
- Daily integration via constant $fETr$ and per-dataset daily reference-ET time series; would require a small post-processing cell and external daily meteorological data.
- Per-class G choice via a landcover-class lookup, restoring a modest version of the original per-class behavior without resurrecting the full two-pass loop.
- Validation cells comparing model output against flux-tower eddy-covariance fluxes for sites that have them.
- An eeMETRIC-style daily-mosaic mode for sites with frequent UAV re-flights.

8 Worked example: WCRC Fruita

8.1 Site and acquisition

The example dataset is a UAV acquisition over an agricultural research facility near Fruita, Colorado (39.167° N, 108.717° W, 1400 m elevation). The flight occurred on 9 August (DOY 221), 2025 at approximately 10:30 AM Mountain Standard Time. Met conditions at the time of acquisition were as follows: air temperature 27.4 °C, wind 1.63 m s^{-1}, vapor pressure 7.2 mbar, incoming shortwave radiation 528 W m^{-2}, atmospheric pressure 1019 mbar. The multiband image (`example_image_clipped.tif`) is a 7-band 0.25 m product covering ~196 × 162 m, with bands 3, 5, and 6 holding red reflectance, NIR reflectance, and thermal brightness temperature respectively.

8.2 Configuration

The dataset config (`Batch_Data/Fruita_Example/config.txt`) selects the alfalfa crop preset with `crop_type=alfalfa` in Section 2 and overrides the canopy height inside the preset block in Section 3.1 (`alfalfa.h_C=0.4` rather than the default 0.5 m, to match the shorter stand observed on the date of the flight). The endmember polygons in `Input/endmember_zones.gpkg` contain one cold polygon (a 460-pixel area inside a well-watered alfalfa field) and one hot polygon (a 288-pixel area on a bare road surface).

A reduced version of the per-flight and per-site sections reads:

```
multiband_image=example_image_clipped.tif
DOY=221
time=10.5
T_A1=27.38
T_A1_units=C
u=1.63
ea=7.2
S_dn=528.25
p=1019
lat=39.16736653032889
lon=-108.71668943835509
alt=1404
crop_type=alfalfa
```

The endmember section is:

```
endmember_mode=zone
endmember_search=2                     # Min/Max (percentile-based)
endmember_zones=./Input/endmember_zones.gpkg
ETrF_bare=0
```

`reference_type=tall` (alfalfa) is the package default, and `G_form=4` uses the alfalfa G/R_n ratio of 0.04.

8.3 Execution performance

End-to-end processing of the 785 × 648 pixel clipped scene completes in roughly five seconds on a late-model laptop. The derivation step writes the four GeoTIFFs (NDVI, FC, LAI, TRAD). The pyMETRIC core itself takes well under a second: the endmember search and dT calibration finish almost instantly, and the image-wide Monin–Obukhov iteration converges in eight iterations on 475 520 valid pixels. The remaining time is spent on LE→ET conversion, reference ET, and zonal stats.

8.4 Primary Outputs

Figure 2 shows the primary model input surfaces generated from the multiband UAV image.The NDVI panel shows alfalfa stands (greens) embedded in a matrix of roads and fallow land. LAI and FC are nonlinear transformations of NDVI through the Beer's-law inversion. TRAD contrasts temperature between the cold alfalfa canopies and hot bare soil areas.

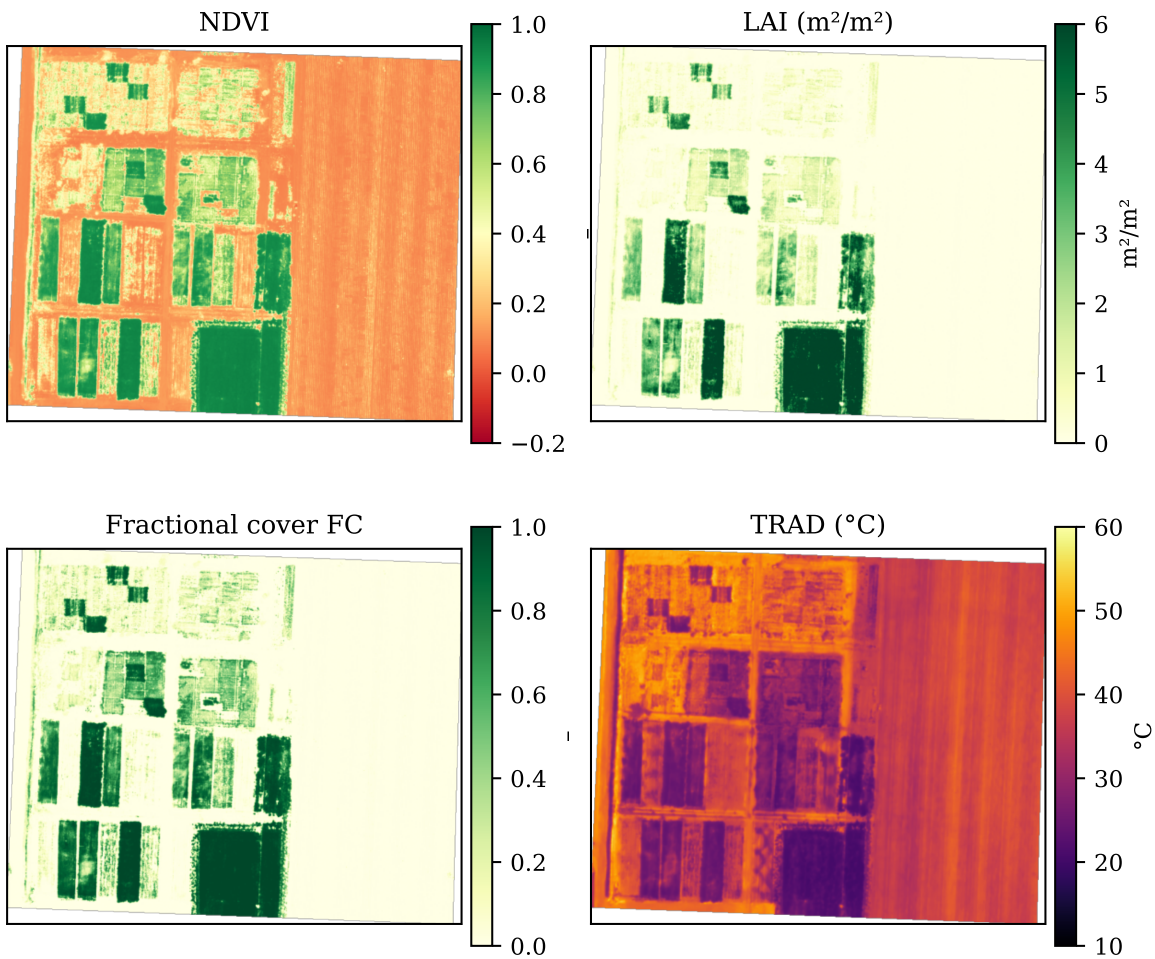

Figure 2: Input rasters derived from the example image.

Figure 3 indicates the user defined locations for hot and cold pixel selection. The delineated cold polygon covers a small section of well-watered alfalfa and the hot polygon covers a section of exposed road.

Figure 4 shows the calibrated linear dT vs T_s relation for Fruita. The intercept $a = -27.7$ K and slope $b = 0.096$ are the same calibration coefficients reported in the processing log. The cold pixel anchors the line at $dT \approx 0.3$ K and the hot pixel at $dT \approx 3.0$ K (because $LE_{\text{hot}} \approx 0$ and most of the available energy becomes sensible heat). On this scene the cold-pixel residual stays positive ($H_{\text{cold}} \approx 13$ W m^{-2}), so the Allen 2013 safeguard (which floors H_{cold} at 5 W m^{-2} when advective demand would otherwise drive it negative) is not triggered here.

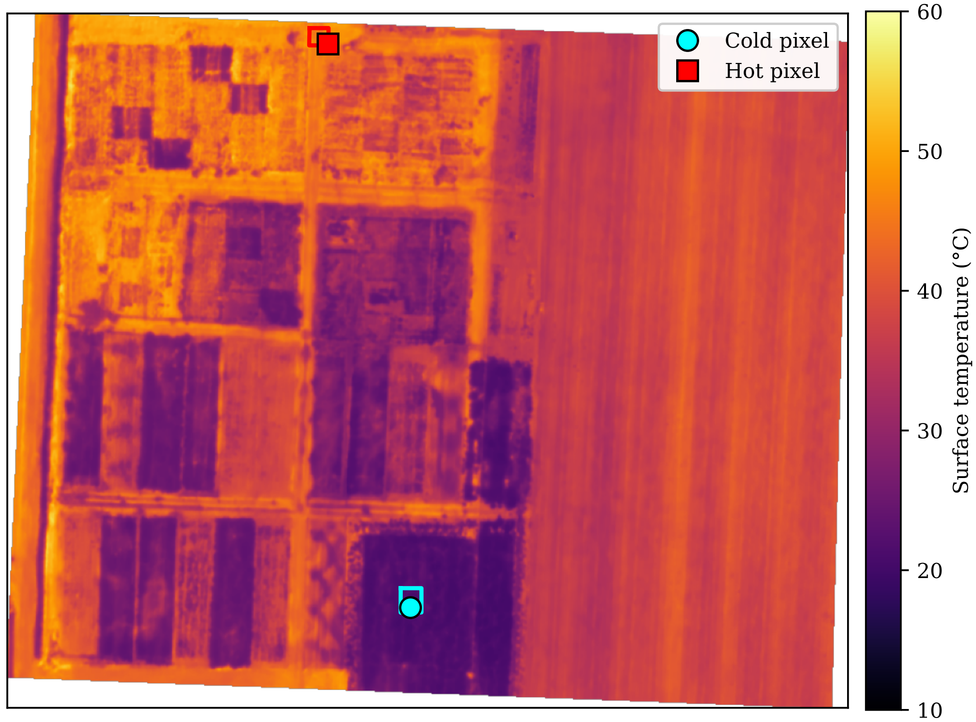

Figure 3: Endmember zone polygons overlaid on TRAD values for the example image.

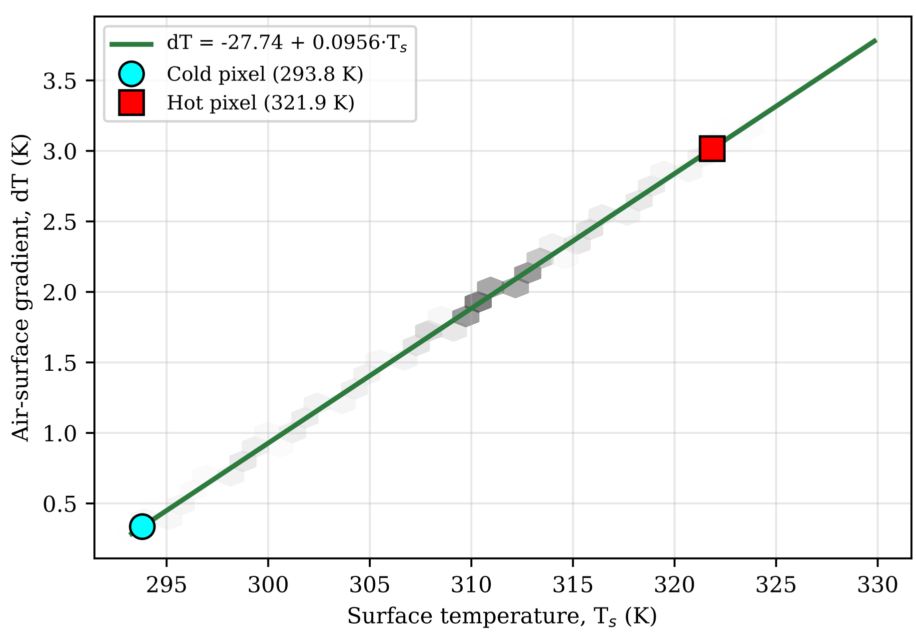

Figure 4: Calibrated dT vs surface temperature for Fruita. The slope and intercept are determined by the two endmember points; the line is then applied image-wide. The grey hexbins indicate the distribution of pixels mapped through the same line, which necessarily falls exactly on it.

The calibration produced

$$dT = -27.74 + 0.0956 \cdot T_s \tag{15}$$

with the cold pixel at row 554, column 376 ($T_s = 293.8$ K, NDVI $= 0.929$) and the hot pixel at row 28, column 299 ($T_s = 321.9$ K, NDVI $= 0.118$). The LST range between endmembers is 28.0 K, well above the 3 K warning threshold. The hot pixel's NDVI of 0.118 falls below the nominal "fully bare" threshold of 0.2, so the search relaxed the VI constraint and accepted the hottest pixel in the hot polygon regardless of VI — an acceptable fallback for this scene given the small bare-soil pool inside the polygon.

Figure 5 shows the distribution of pixel-wise instantaneous ET. The distribution exhibits heavy tails. a dominant primary mode near 0.2 mm hr^{-1} represents partially-vegetated and bare-soil pixels (the bulk of the scene). A heavy tailing out to ~0.5 mm hr^{-1} reflect the influence of well-watered alfalfa canopy.

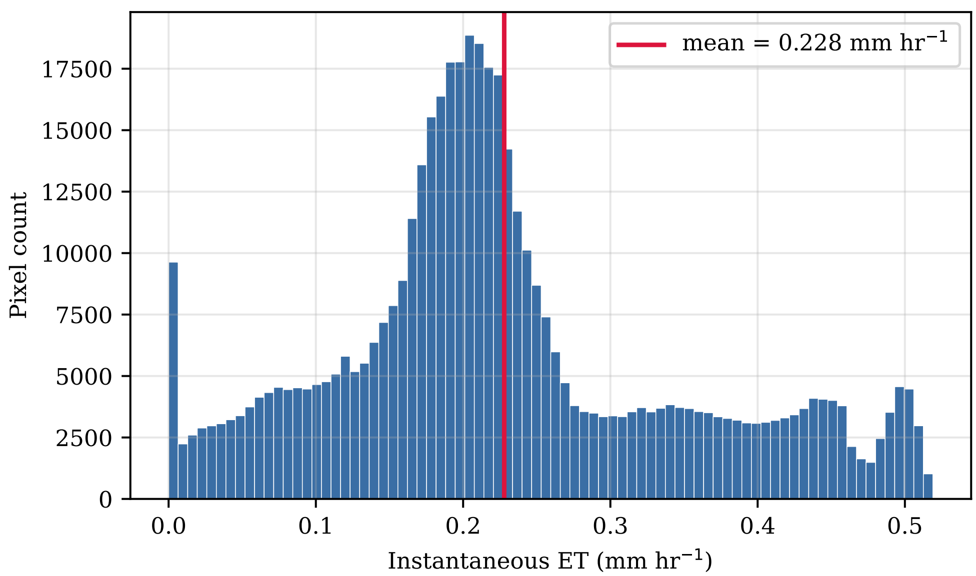

Figure 5: Distribution of pixel-wise instantaneous ET across the example image.

Figure 6 shows the four energy balance components for the example image scene. The well-watered alfalfa visibly dominates the LE panel, while H peaks over the exposed roads and gravel pad. R_n is dominated by net shortwave (large over high-albedo bare soil but also large over the dark canopy with low albedo), and the partitioning between H and LE shifts sharply at the canopy/soil boundary.

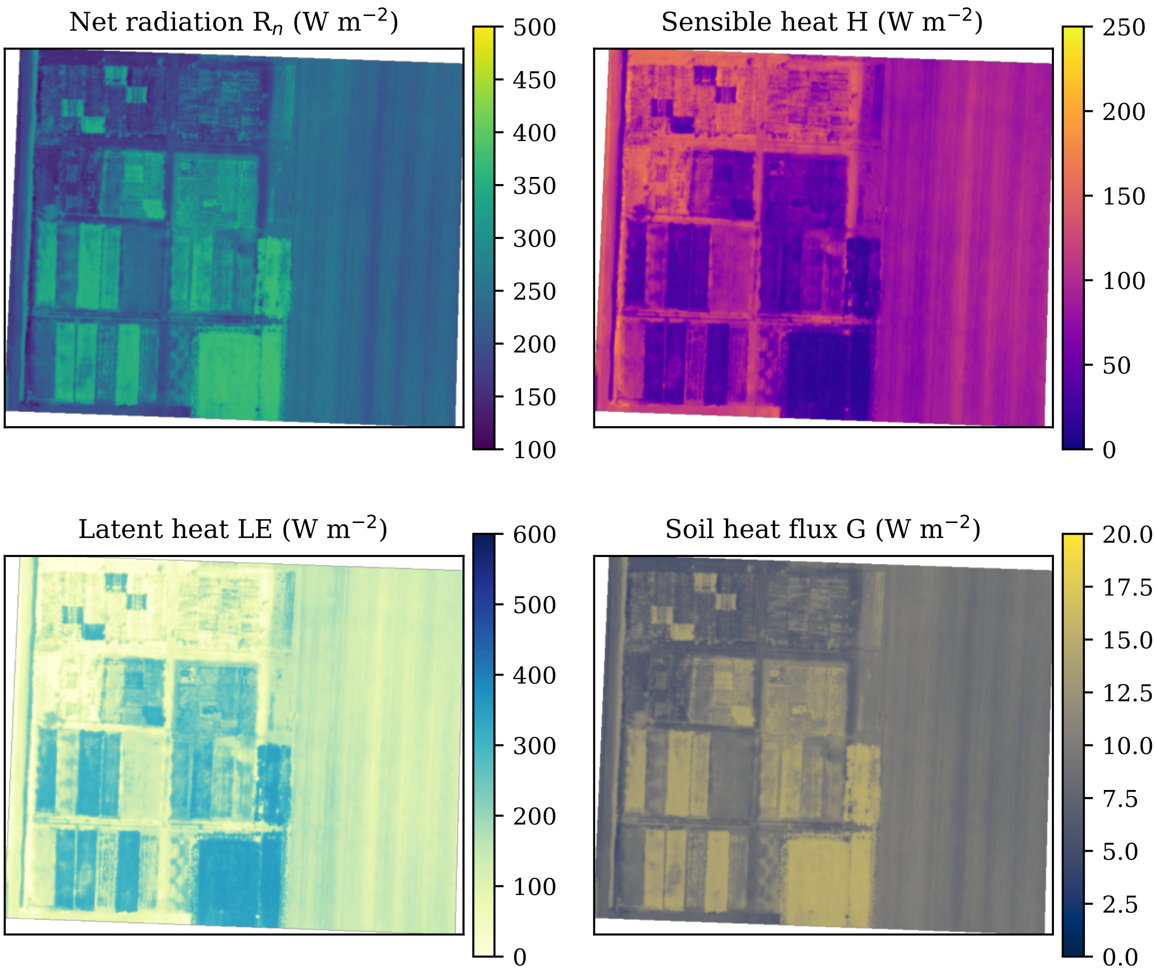

Figure 6: Instantaneous surface energy balance components for the example image.

Figure 7 displays an ET map with the zonal-statistics plot polygons overlaid. ET is highest in the well-watered alfalfa plots (~0.3–0.5 mm hr^{-1}, dark blue) and approaches zero in the adjacent bare soil and roads (very pale, near white).

Figure 8 shows the same scene expressed as ET fraction against the tall (alfalfa) reference. Values near 1.0 indicate the surface is transpiring close to its reference potential. Values near zero indicate bare soil or water stress.

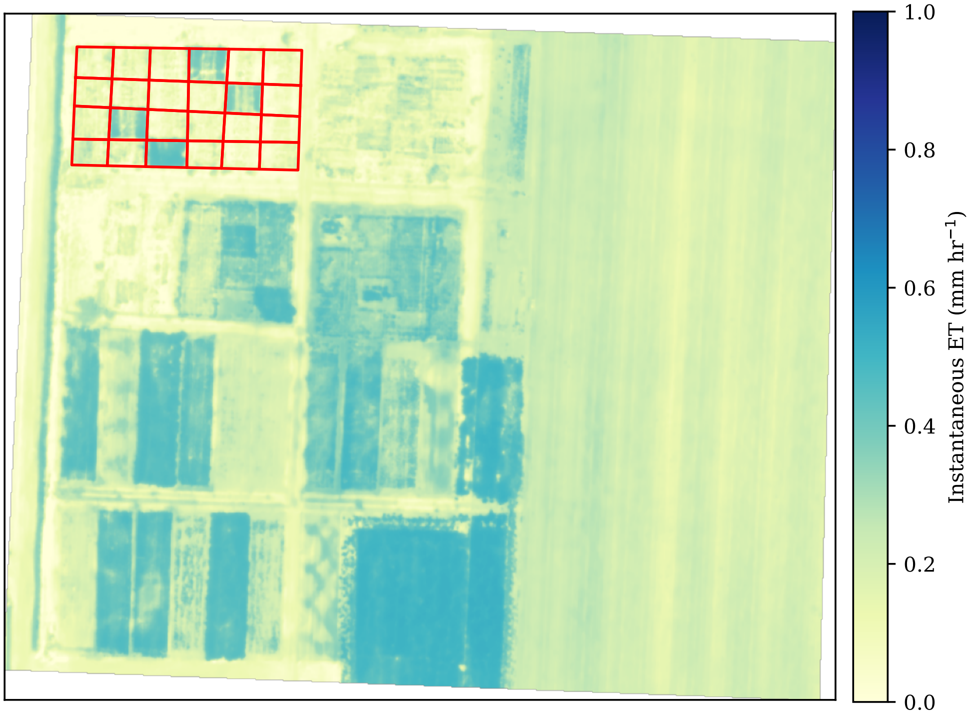

Figure 7: Instantaneous ET (mm hr^{-1}) over the example image, with the study-plot polygons overlaid.

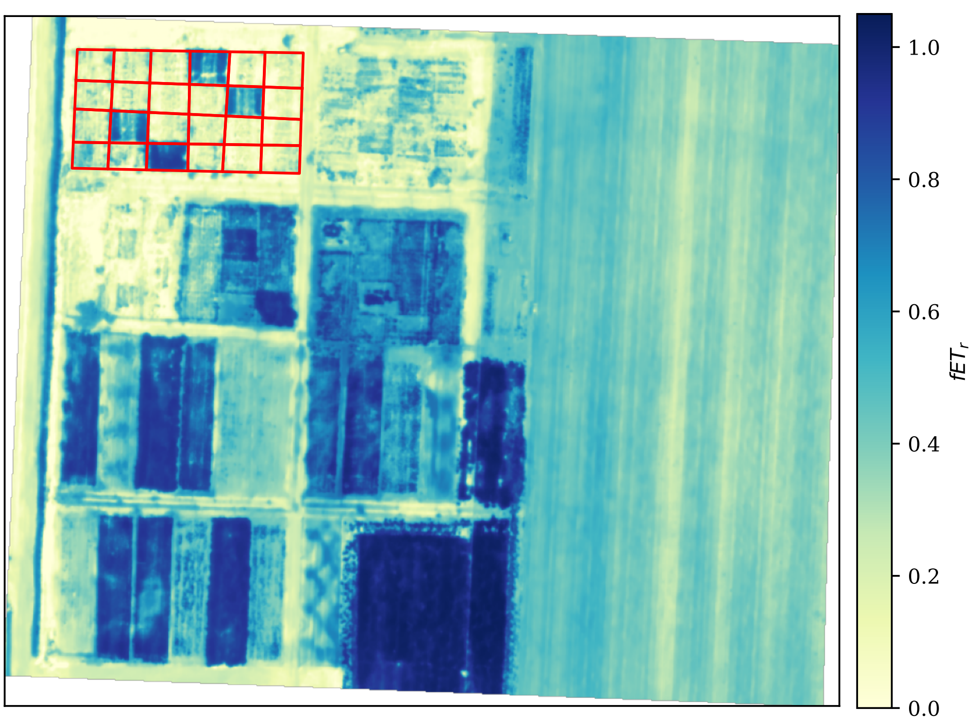

Figure 8: ET fraction ($fETr = LE/ET_{ref}$) for the example image.

Figure 9 shows the per-plot mean ET ± 1 SD for the example study plots. Plot-to-plot variability reflects differences in canopy density, irrigation uniformity, and potential edge effects from adjacent surfaces. The plots with the highest mean ET (~0.30–0.40 mm hr^{-1}, plots 5, 10, 15, and 24) correspond to the densest crop stands; the plots with the lowest means (~0.05 mm hr^{-1}, plots 1, 6, 12, 13, and 19) are sparser or include substantial bare-soil patches within the plot boundary.

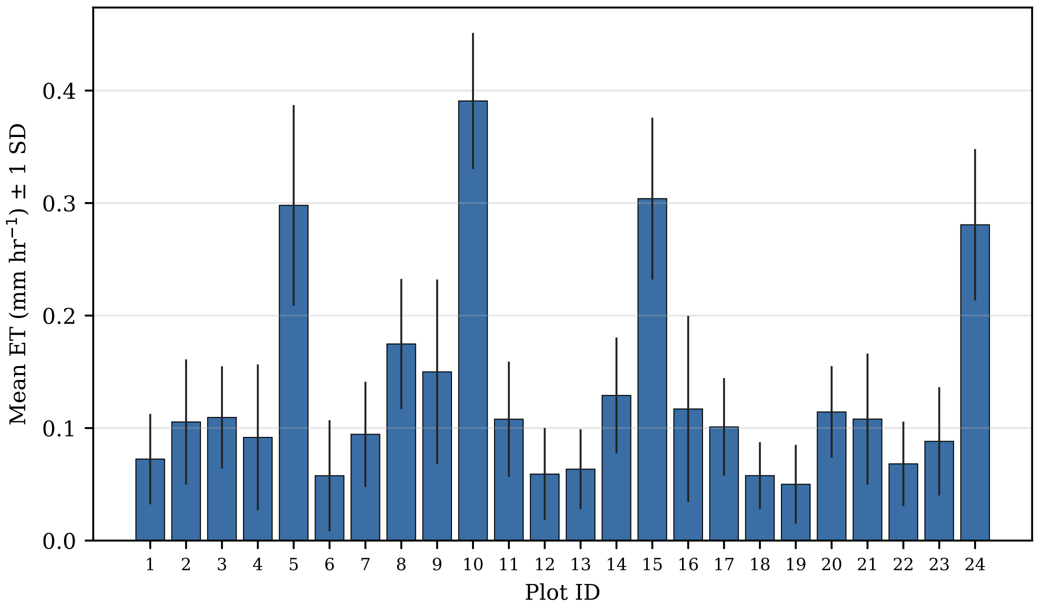

Figure 9: Per-plot instantaneous ET means (± 1 SD) for the example study plots.

Image-wide statistics for the energy-balance components are summarized in Table 7. The computed reference ET for the same meteorological conditions is $ET_r = 0.527$ mm hr^{-1} (alfalfa). The mean ET / ET_r ratio of ~0.43 reflects a scene dominated by bare soil and roads with isolated patches of fully-closed crop canopy. The densest alfalfa pixels themselves reach $fETr \approx 1.0$.

Table 7: Image-wide statistics for the example image

Variable	Mean	Median	Min	Max
R_n (W m^{-2})	247	237	100	380
H (W m^{-2})	84	88	11	197
LE (W m^{-2})	153	140	-102	354
G (W m^{-2})	10	9	4	15
ET (mm hr^{-1})	0.228	0.209	-0.154	0.519
$fETr$ (vs alfalfa)	0.45	0.41	-0.30	1.04

A small fraction (~1.6%) of pixels show slightly negative LE (and consequently negative ET) where modeled H exceeds $R_n - G$ on the very hottest bare patches. These can be floored at zero in post-processing for reporting, but are left as-computed here to make the energy-balance closure inspectable.

8.5 Cross-strategy sensitivity analysis

Figure 10 shows results from the sensitivity analysis for the example image. The five hot/cold pixel search strategies produce ET values that span 0.177–0.235 mm hr^{-1}. The ~25% relative spread underscores the sensitivity of internal calibration to anchor choice. `MinMax_auto` (0.235 mm hr^{-1}) and `MinMax_zone` (0.228 mm hr^{-1}) agree to within 3%, indicating that the zone polygons and the full-image percentile search converge on essentially the same cold and hot anchors. `CIMEC_zone` (0.221 mm hr^{-1}) is similarly close. `ESA_auto` (0.201 mm hr^{-1}) and `CIMEC_auto` (0.177 mm hr^{-1}) sit at the low end. When run without zone constraints, the CIMEC algorithm's tight albedo filter pushes the cold anchor away from the best-watered pixel in this scene.

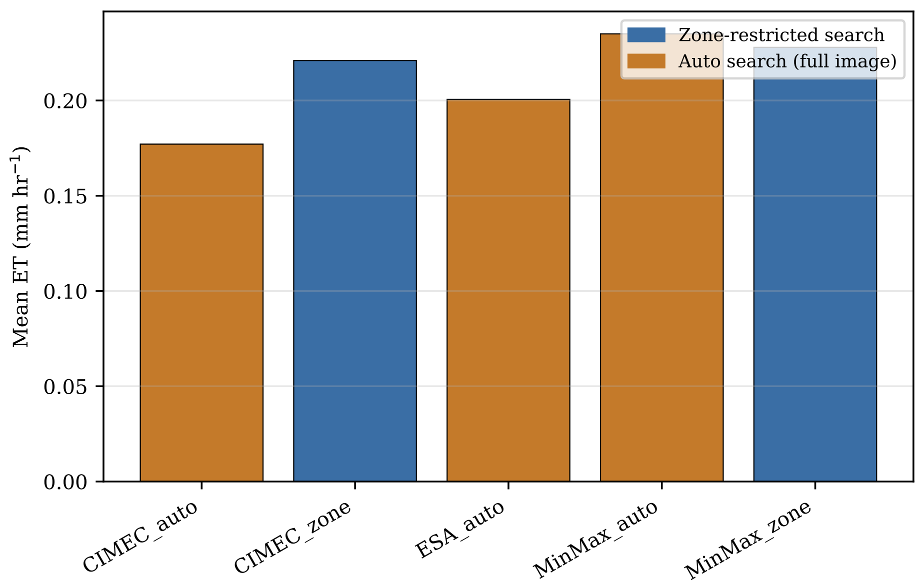

Figure 10: Image-wide mean ET under five endmember strategies for the example image.

The calibration anchors selected by each strategy, the resulting dT line, and the image-mean ET are summarized in Table 8. The cold pixel is essentially the same across strategies ($T_{\text{cold}} \approx 293.7$–294.2 K, $VI \approx 0.92$–0.94) — every strategy locates the same well-watered alfalfa cluster. The hot pixel varies more: `MinMax_auto` picks the hottest pixel in the scene (322.6 K), while `CIMEC_auto` settles on a noticeably cooler hot anchor (316.95 K) because its tight albedo filter rejects the brightest pixels. A smaller LST range gives a steeper dT slope (b), which converts more pixel-level T_s variation into H and therefore leaves less energy for LE — that mechanism is what drives `CIMEC_auto`'s steeper slope and lower image-mean ET. On this scene, the cold-pixel residual stays positive under every strategy, so the Allen 2013 safeguard is not triggered and the calibration is set by the energy balance at the selected anchors rather than by the floor.

Table 8: Endmember anchors (raw surface temperatures) and calibrated $dT = a + bT_s$ for each search strategy on the example image. The Allen 2013 safeguard was not triggered in any strategy on this scene.

Strategy	T_{cold} (K)	T_{hot} (K)	ΔT (K)	a (K)	b	Mean ET (mm hr ^{1})
`MinMax_zone`	293.80	321.85	28.0	−27.74	0.0956	0.228
`MinMax_auto`	293.76	322.60	28.8	−27.56	0.0947	0.235
`CIMEC_zone`	294.22	320.17	26.0	−30.79	0.1057	0.221
`CIMEC_auto`	294.09	316.95	22.9	−40.38	0.1385	0.177
`ESA_auto`	293.65	319.30	25.7	−34.27	0.1178	0.201

9 Validation against WSWUP/pyMETRIC

To confirm that the adapted solver still reproduces the canonical METRIC solution, it is cross-checked here against code developed by the Desert Research Institute (DRI) Western States Water Use Program (WSWUP) and housed in the `WSWUP/pyMETRIC` GitHub repository. The WSWUP implementation is widely used for operational Landsat image analysis and serves as a *de facto* reference for the METRIC algorithm (Allen, Tasumi, and Trezza 2007). The `pyMETRIC-UAV` solver was validated against the `WSWUP/pyMETRIC` solver using the Fruita worked-example scene and a common set of meteorological inputs.

`WSWUP/pyMETRIC` and `pyMETRIC-UAV` derive their inputs differently. `WSWUP/pyMETRIC` computes surface variables from Landsat bands and ingests gridded reference ET, whereas `pyMETRIC-UAV` consumes user NDVI/LST rasters and computes reference ET internally. A meaningful comparison between the two code bases required isolation of the energy-balance solvers in the respective code bases. To do this, the `WSWUP/pyMETRIC` energy-balance functions (`et_numpy.py`) were driven through a pixel-scale calibration loop and raster-scale stability loop. Identical net radiation R_n, soil heat flux G, surface temperature T_s, momentum roughness z_{0m}, and the same two anchor pixels, were provided to each solver. Each solver then performed its own dT calibration, aerodynamic-resistance calculation, and Monin–Obukhov stability iteration. Any disagreement in the final solutions were assumed attributable to the solvers rather than derivation of input coverages and variables.

The two solvers recovered a similar dT calibration. The `WSWUP/pyMETRIC` solver returned a near-surface-gradient slope of 0.102 (intercept −29.5 K) against `pyMETRIC-UAV's` slope of 0.096 (intercept −27.7 K). Instantaneous ET computed by `pyMETRIC-UAV` tracked the `WSWUP/pyMETRIC` reference with $R^2 = 0.999$ and an RMSE of 0.004 mm hr^{-1}, and the image-mean ET was essentially identical between the two solvers (0.228 mm hr^{-1} for `pyMETRIC-UAV` versus 0.227 for the `WSWUP/pyMETRIC` reference; Table 9). Sensible and latent heat agreed to within an RMSE of 0.6 W m^{-2}.

Table 9: Solver cross-validation on the Fruita scene (n = 475,520 pixels). MBE and OLS slope are pyMETRIC-UAV relative to the WSWUP reference (MBE = UAV − WSWUP).

Variable	R^2	RMSE	MBE	OLS slope
H (W m^{-2})	1.000	0.64	−0.48	0.993
LE (W m^{-2})	1.000	0.64	+0.48	0.997
ET (mm hr^{-1})	0.999	0.004	+0.001	0.990
r_{ah} (s m^{-1})	0.972	2.91	+2.69	0.594

The closest agreement is in the fluxes that METRIC self-calibration constrains most directly. Because both solvers are pinned to the same two anchor pixels and are driven with identical T_s, R_n, and G, the spatial H field is largely determined. Latent heat and ET follow as the energy residual, so their image means coincide to better than 1%.

The one term where the solvers genuinely differ is the aerodynamic resistance r_{ah}, which shows the weakest agreement ($R^2 = 0.972$, OLS slope 0.59). The UAV model's r_{ah} runs somewhat higher than the reference (image means 26.7 versus 24.0 s m^{-1}). This is the expected consequence of two different aerodynamic treatments. `pyMETRIC-UAV` iterates r_{ah} with pyTSEB's Monin–Obukhov scheme using the measured 2 m wind speed over each pixel's own roughness, whereas the WSWUP code extrapolates wind speed out to a height of 200 m using a fixed roughness value. Crucially, this difference is absorbed by the self-calibration rather than propagating into ET. METRIC inverts the anchors through r_{ah} to fit a fixed anchor H. A solver with different r_{ah} simply compensates with a slightly different dT and the resulting ET values stay within a fraction of a percent in the mean.

This validation check establishes solver equivalence on a single UAV scene, not agreement between the different data processing chains used by `pyMETRIC-UAV` and `WSWUP/pyMETRIC`. Within those bounds, `pyMETRIC-UAV` solves the same METRIC energy balance as `WSWUP/pyMETRIC`. This finding confirms that the UAV-specific changes documented above alter how the hot and cold pixels are selected, not the physics of the underlying METRIC solution.

10 Conclusion

`pyMETRIC-UAV` adapts the METRIC surface-energy-balance model to sub-meter UAV thermal and multispectral imagery while preserving the algorithmic core that gives METRIC its operational robustness. The single-pass refactor, explicit `reference_type` abstraction, four-mode endmember system, zone-coupled NDVI calibration, resolution-adaptive spatial filtering, and percentile-based Min/Max selector together close the most common failure modes of the upstream versions of the `pyMETRIC` code when run on raw UAV imagery. The configuration is organized around a Jupyter notebook that gets a non-coder from a fresh dataset to a per-pixel ET map in roughly ten minutes of edit time and less than thirty seconds of compute time. The package is in active use for plot-level water-use studies in irrigated agriculture and is available at the project repository.

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A Appendix: Configuration key reference

The following table lists every key recognized by `pyMETRIC.METRICConfigFileInterface`, grouped by section. Defaults shown in parentheses; "—" means no default (key must be supplied).

A.1 Flight info

A.1.1 Input image + sensor band assignments

Key	Default	Description
`multiband_image`	—	Filename of multiband UAV image (in dataset folder)
`red_band`	—	Red band index (1-based)
`nir_band`	—	NIR band index
`thermal_band`	—	Thermal band index
`thermal_units`	C	C or K

A.1.2 Date and time of acquisition

Key	Default	Description
`DOY`	—	Day of year (1–366)
`time`	—	Decimal hours, local standard time

A.1.3 Meteorology

Key	Default	Description
`T_A1`	—	Air temperature (units below)
`T_A1_units`	K	C=Celsius, K=Kelvin
`u`	—	Wind speed (m s^{-1})
`ea`	—	Water-vapor pressure (mbar)
`S_dn`	—	Incoming shortwave (W m^{-2})
`p`	—	Atmospheric pressure (mbar)
`L_dn`	(estimate)	Incoming longwave (W m^{-2}); blank → estimate from T_a, e_a

A.1.4 Polygon file paths

Key	Default	Description
endmember_zones	—	Polygon file with cold/hot features (dual-purpose: NDVI calibration + endmember search)
zonal_polygons	—	Polygon file for plot-level zonal stats
input_mask	0	Processing mask raster (0 = all valid pixels)

A.1.5 Endmember search

Key	Default	Description
endmember_mode	auto	auto, zone, manual, prescribed
endmember_search	0	0=CIMEC, 1=ESA, 2=Min/Max

ETrF tuning (ETrF_bare, hot_ETrF) is in Section 3.8. Mode-specific extras (manual pixel coords, prescribed dT, bounding-box fallback, zone-field override) are in Section 3.9.

A.2 Site info

Key	Default	Description
lat	—	Site latitude (decimal degrees, N+)
lon	—	Site longitude (decimal degrees, E+)
alt	—	Elevation above sea level (m)
stdlon	—	Time-zone standard meridian (degrees)
z_T	—	Air-temperature measurement height (m AGL)
z_u	—	Wind-speed measurement height (m AGL)
crop_type	alfalfa	Selects which preset in Section 3.1 is active. Built-ins: alfalfa, corn, wheat, pasture, orchard
reference_type	tall	tall = alfalfa (ETr), short = grass (ETo)

A.3 Advanced settings

A.3.1 Crop presets

Preset-specific entries use a <crop_type>.<key> namespace (e.g. alfalfa.k_ext=0.6). Only the block matching the active crop_type is promoted to the flat namespace at parse time.

Key (per preset)	Range	Description
`<crop>.k_ext`	0.4–0.7	Canopy light extinction coefficient
`<crop>.h_C`	0.1–10 m	Canopy height
`<crop>.landcover`	IGBP class	12=Crop, 10=Grass, 4=Broadleaf-D, etc.
`<crop>.NDVI_full`	0.7–0.95	NDVI of fully vegetated surface (overridden by zone calibration if available)
`<crop>.rho_vis_C`	0.05–0.12	Leaf visible reflectance
`<crop>.tau_vis_C`	0.04–0.10	Leaf visible transmittance
`<crop>.rho_nir_C`	0.30–0.50	Leaf NIR reflectance
`<crop>.tau_nir_C`	0.20–0.40	Leaf NIR transmittance

A.3.2 Site-level canopy / leaf parameters (apply to all crops)

Key	Default	Description
`w_C`	1.0	Canopy width:height ratio
`x_LAD`	1.0	Leaf angle distribution (1=spherical)
`emis_C`	0.98	Leaf emissivity
`NDVI_soil`	0.15	NDVI of bare soil at this site (overridden by zone calibration)
`LAI_max`	8.0	Upper clamp on derived LAI

A.3.3 Soil heat flux

Key	Default	Description
`G_form`	4	0=const, 1=ratio, 2=diurnal, 4=ASCE, 5=Allen empirical
`G_tall`	0.04	G/R_n for alfalfa (used when `reference_type=tall`)
`G_short`	0.10	G/R_n for grass (used when `reference_type=short`)
`G_constant`	0	for G_form=0
`G_ratio`	0.35	for G_form=1
`G_amp`	35	for G_form=2
`G_phase`	3.5	for G_form=2
`G_shape`	2.0	for G_form=2

A.3.4 Quality control

Key	Default	Description
`min_lst_range`	3.0 K	LST range warning threshold between endmembers

Key	Default	Description
cv_window_meters	5.0 m	Spatial homogeneity filter window (auto-scaled to pixels)

A.3.5 Soil spectral properties

Key	Default	Description
emis_S	0.95	Soil emissivity
rho_vis_S	0.15	Soil visible reflectance
rho_nir_S	0.25	Soil NIR reflectance

A.3.6 Aerodynamic resistance

Key	Default	Description
use_METRIC_resistance	1	0=Kustas-Norman 1999, 1=Allen 1998
z0_soil	0.01 m	Bare-soil roughness length

A.3.7 Optional file path overrides

Key	Default	Description
T_R1, VI, LAI, f_c	(auto-discovered from Input/)	Pre-computed raster paths
output_file	(auto from folder name)	Primary output path

A.3.8 Hot pixel ETrF tuning

Key	Default	Description
ETrF_bare	0	ETrF for fully bare soil (0–1 or raster path)
hot_ETrF	—	Direct override of hot-pixel ETrF

A.3.9 Alternate endmember specifications

Key	Default	Description
endmember_zone_field	zone	Attribute field for cold/hot label in the zones polygon file

Key	Default	Description
`cold_zone`, `hot_zone`	—	Bounding-box fallback `xmin,ymin,xmax,ymax` (used when no polygon file)
`cold_pixel_rc`, `hot_pixel_rc`	—	Manual mode: pixel coords
`cold_pixel_xy`, `hot_pixel_xy`	—	Manual mode: geographic coords
`prescribed_dT_a`, `prescribed_dT_b`	—	Prescribed mode: dT line coefficients

B Appendix: Symbol glossary

Symbol	Meaning	Units
R_n	Net radiation at the surface	W m^{-2}
R_{ns}	Net shortwave radiation	W m^{-2}
R_{nl}	Net longwave radiation	W m^{-2}
H	Sensible heat flux	W m^{-2}
LE	Latent heat flux	W m^{-2}
G	Soil heat flux	W m^{-2}
T_s	Surface temperature (TRAD)	K
$T_{s,\mathrm{datum}}$	Surface temperature adjusted to common elevation datum	K
T_a	Air temperature	K
dT	Air-surface temperature gradient	K
u	Wind speed	m s^{-1}
u_2	Wind speed at 2 m height	m s^{-1}
u_*	Friction velocity	m s^{-1}
e_a	Actual vapor pressure	mbar or kPa
e_s	Saturation vapor pressure	mbar or kPa
p	Atmospheric pressure	mbar
α	Broadband shortwave albedo	—
ε	Broadband surface emissivity	—
ρ	Air density	kg m^{-3}
c_p	Heat capacity of moist air	J kg^{-1} K^{-1}
r_{ah}	Aerodynamic resistance to heat transfer	s m^{-1}
L	Monin–Obukhov length	m
γ_w	Moist adiabatic lapse rate	K m^{-1}
γ	Psychrometric constant	kPa $°C^{-1}$
Δ	Slope of saturation vapor pressure curve	kPa $°C^{-1}$
λ	Latent heat of vaporisation	J kg^{-1}
σ	Stefan–Boltzmann constant	W m^{-2} K^{-4}
z	Elevation above sea level	m
z_{0m}	Aerodynamic roughness length for momentum	m
d_0	Zero-plane displacement height	m
h_C	Canopy height	m
w_C	Canopy width:height ratio	—
f_c	Fractional vegetation cover	—

Symbol	Meaning	Units
LAI	Leaf area index	m^2 m^{-2}
k_{ext}	Canopy light extinction coefficient	—
$NDVI$	normalized difference vegetation index	—
ET_{ref}	ASCE reference ET (energy units)	W m^{-2}
ET_r	ASCE tall reference ET (mass units)	mm hr^{-1}
ET_o	ASCE short reference ET (mass units)	mm hr^{-1}
ET_{sz}	ASCE standardized reference ET (Equation 10); equals ET_r or ET_o per `reference_type`	mm hr^{-1}
$fETr$	ET fraction LE/ET_{ref}	—
C_n, C_d	ASCE reference-ET equation constants	—
a, b	Calibrated dT intercept and slope	K, —